deliverables\final-checklist.md

Final Submission Checklist

Submission Platform

- [ ] Project created on Microsoft Innovation Studio Hackathon platform
- [ ] Project description pasted and reviewed
- [ ] Public GitHub repository link added
- [ ] Demo video link added
- [ ] Architecture image uploaded
- [ ] Screenshots / GIFs uploaded

Repo Readiness

- [ ] Top-level `README.md` reflects the current unified app architecture
- [ ] Setup instructions are accurate
- [ ] Azure deployment path is accurate
- [ ] Known limitations are honest and concise
- [ ] Latest working code is committed

Required Narrative Assets

- [ ] `deliverables/submission-summary.md` finalized
- [ ] `deliverables/workflows.md` finalized
- [ ] architecture diagram finalized
- [ ] `deliverables/demo-outline.md` finalized

Demo Video

- [ ] 2 minutes or less
- [ ] shows review -> create -> Excel flow
- [ ] shows reactive recalculation
- [ ] includes reverse Excel -> review flow if time allows
- [ ] audio is understandable
- [ ] final file/link tested in browser

Screenshots / GIFs

- [ ] unified app home / sidebar
- [ ] reviewed script table
- [ ] created article section with mode links
- [ ] Excel loaded PQ table
- [ ] Excel reactive recalculation update
- [ ] optional GIF for bridge flow

Final Verification

- [ ] local app works with Excel bridge
- [ ] cloud deployment works in browser
- [ ] sample prompt executes successfully
- [ ] created article opens in CalcsLive
- [ ] all links in submission are valid

Deadline Safety

- [ ] Aim to finalize before March 15, 11:59 PM Pacific Time
- [ ] Treat March 15 as submission buffer, not core build time

deliverables\workflows.md

Supported Workflows

This document defines the supported end-user workflows for the CalcsLive Agent submission. It is the source of truth for demo planning, README polish, screenshots, and submission copy.

Product Scope

CalcsLive Agent is a moderator-style AI application that connects:

- CalcsLive as the unit-aware calculation engine and reusable article store
- a unified Azure-hosted / local Streamlit app as the main user interface
- a local Excel bridge for spreadsheet interaction when Excel is available

The same unified app supports both cloud-only and local-with-Excel modes.

Modes

Cloud Mode

Available when the app is running without access to the local Excel bridge.

Capabilities:

- create and review stateless calculation scripts
- discover units / resolve ambiguous unit aliases through CalcsLive tools
- persist reviewed scripts as CalcsLive calculation articles
- open created calculations in multiple CalcsLive modes (`edit`, `calculate`, `table`, `view`)

Local + Excel Mode

Available when the local Excel bridge is reachable.

Capabilities:

- all cloud capabilities
- send created calculations to Excel
- read calculations from Excel back into reviewed CalcsLive script form
- reactive recalculation in Excel after edits

Workflow 1: Review -> Create -> Send to Excel

This is the primary showcase workflow.

User goal

Create a new engineering calculation from natural language, review the generated script, persist it, and then use it interactively in Excel.

Steps

1. User enters a natural-language prompt in the unified app.
2. Agent interprets the problem and generates a PQ script.
3. Agent runs `run_calcslive_script` for stateless review.
4. User reviews:
    - generated title
    - description
    - calculation table
    - warnings

    - category metadata
5. User clicks `Create Article`.
6. App persists the article using `create_calcslive_article_from_script`.
7. If Excel is available, user clicks `Send Calc to Excel`.
8. Bridge loads the article into Excel as metadata + PQ table.

Outcome

The user now has a reusable CalcsLive article and a structured Excel version of that calculation.

Workflow 2: Edit in Excel -> Reactive Recalculation

This demonstrates closed-loop spreadsheet interaction.

User goal

Modify input values or units directly in Excel and see outputs update automatically.

Steps

1. A calculation is already loaded into Excel.
2. User edits input values and/or units in the spreadsheet.
3. Excel bridge detects sheet changes through COM events.
4. Bridge reads the current PQ table.
5. Agent/bridge sends updated inputs to CalcsLive.
6. Returned outputs are written back to Excel.

Outcome

Excel behaves like a unit-aware engineering workspace while CalcsLive remains the source of truth for the computational logic.

Workflow 3: Excel-authored Table -> Review -> Create

This demonstrates reverse flow from spreadsheet to reusable CalcsLive content.

User goal

Start from a manually prepared Excel PQ table, review it as a CalcsLive script, and turn it into a persistent calculation article.

Expected Excel structure

The bridge looks for a compatible table with these semantic columns:

- `Description`
- `Symbol`
- `Expression`
- `Value`
- `Unit`

Optional:

- left-side row number column (#)

Row rules

- Input row:
    - `Expression` blank
    - `Value` and `Unit` populated
- Output row:
    - `Expression` populated

- `Unit` populated
- `Value` optional or pre-existing

Steps

1. User lays out a compatible PQ table in Excel.
2. User clicks `Get Calc from Excel`.
3. Bridge reads the Excel table and converts it into `pqs`, `inputs`, and `outputs`.
4. App runs `run_calcslive_script` for review.
5. User adjusts title/description if needed.
6. User clicks `Create Article`.

Outcome

An Excel-authored calculation becomes a persistent, reusable CalcsLive article.

Workflow 4: Cloud-only Review and Persistence

This is the fallback / remote-friendly path for users without Excel.

User goal

Create and persist new CalcsLive calculations entirely in the cloud app.

Steps

1. User describes the desired calculation.
2. Agent generates and reviews the script using `run_calcslive_script`.
3. User validates warnings, units, expressions, and output structure.
4. User clicks `Create Article`.
5. User opens the result in CalcsLive through one of the generated mode links.

Outcome

The app works as a standalone cloud-native CalcsLive creation interface even when no Excel bridge is present.

User Mental Model

- `Reviewed CalcsLive Script` = human-in-the-loop validation stage
- `Create Calculation Article` = persistence stage
- `Bridge to/from Excel` = interoperability stage

This ordering is intentional and reflects the system's moderator-agent design.

What Is Demonstrated Best In Submission

Recommended emphasis order:

1. Review -> Create -> Send to Excel
2. Edit in Excel -> Reactive recalculation
3. Excel-authored table -> Review -> Create
4. Cloud-only review/create flow

Current Non-Goals / Deferred Work

- automatic reconnect from cloud app directly to a private local Excel bridge
- full Excel MCP parity for the complete review/create lifecycle
- persisted override/edit API for existing saved article instances
- generalized plugin system beyond Excel (future path: CAD, Sheets, others)

deliverables\submission-summary.md

Submission Summary Draft

Project Name

CalcsLive Agent - Unit-aware Calculation Carrier for Excel in MAF

One-line Summary

An Azure-orchestrated moderator agent that turns spreadsheets into a bi-directional, unit-aware engineering workflow using CalcsLive and Excel.

Problem

Excel is one of the most common tools for engineering and technical calculations, but it is fundamentally unit-blind. Users can easily mix inches, millimeters, kilograms, liters, and custom engineering units in the same spreadsheet without any native guardrails. This creates avoidable risk, rework, and calculation errors.

At the same time, many technical calculations begin in one place and need to move across multiple contexts: reviewed by a user, persisted as reusable logic, loaded into a spreadsheet, edited interactively, and recalculated reliably.

Solution

CalcsLive Agent makes CalcsLive the unit-aware calculation engine behind a moderator-style AI workflow.

The app supports a review-first lifecycle:

1. A user describes a calculation in natural language.
2. The agent generates and reviews a CalcsLive-compatible PQ script.
3. The user validates warnings, units, and expressions before persistence.
4. The reviewed script is persisted as a reusable CalcsLive article.
5. When Excel is available, the same article can be sent into Excel, edited there, and recalculated reactively.

The system also supports the reverse direction:

- a compatible Excel table can be read back into the agent,
- converted into a reviewed CalcsLive script,
- and persisted as a new reusable article.

This creates a two-way bridge between spreadsheet work and reusable, unit-aware calculation definitions.

Why It Matters

- reduces spreadsheet risk from mixed-unit calculations
- allows human-in-the-loop review before persistence
- makes calculations reusable across interfaces instead of trapped in one workbook
- demonstrates how AI agents can moderate engineering workflows rather than just generate text

Core Features

- natural-language calculation generation
- review-first stateless script execution
- persistence of reviewed scripts as CalcsLive articles

- multiple article modes (`edit`, `calculate`, `table`, `view`)
- Excel bridge for loading structured calculation tables into spreadsheets
- reactive recalculation after Excel edits
- reverse flow from Excel-authored PQ table back into CalcsLive review/create workflow

Microsoft Technologies Used

- Azure-hosted deployment via Azure Container Apps
- Azure AI / Microsoft AI platform-backed agent orchestration in the app
- Microsoft Agent Framework-aligned moderator/orchestration architecture
- Azure MCP / MCP-oriented extensibility story through toolized local and API integrations
- GitHub + VS Code development workflow

Architecture Summary

The project uses a unified Streamlit moderator app as the main user entrypoint.

- In cloud mode, the app supports review and persistence of CalcsLive scripts.
- In local mode, the same app detects the Excel bridge and enables bi-directional spreadsheet workflows.
- CalcsLive acts as the source-of-truth unit-aware engine and article store.
- Excel acts as a familiar interactive workspace for engineering calculations.

Best-Fit Categories

Primary:

- Build AI Applications & Agents

Strong secondary alignment:

- Best Multi-Agent System
- Best Azure Integration

Differentiator

This project is not just “AI inside Excel.” It demonstrates a moderator-agent workflow where reviewed, unit-aware calculation logic can move between CalcsLive and Excel in both directions. The result is a system-level unit-awareness story for spreadsheets with reusable calculation definitions.

Known Limits / Honest Scope

- cloud deployment does not automatically connect to a private local Excel bridge
- full Excel MCP parity is not required for the working demo path
- some advanced persistence/edit workflows are intentionally deferred after hackathon

What Reviewers Should See in the Demo

1. Generate and review a new calculation
2. Create the article
3. Send it to Excel
4. Change Excel inputs and show reactive recalculation
5. Read an Excel-authored table back into the app and persist it as a new article

`deliverables\demo-outline.md`

Final Demo Outline

Target Length

2 minutes maximum

Goal

Show a clear, working story that combines:

- review-first AI calculation generation
- persistence into CalcsLive
- Excel interoperability
- reactive unit-aware spreadsheet recalculation

Recommended Demo Flow

0:00 - 0:15 | Problem framing

Say:

“Excel is widely used for engineering calculations, but it is unit-blind. CalcsLive Agent turns Excel into a bi-directional, unit-aware engineering workflow.”

Show:

- unified app home screen

0:15 - 0:35 | Review-first calculation creation

Action:

- paste a sample prompt

Show:

- reviewed CalcsLive script
- generated title/description
- calculation table
- warnings or category metadata if present

Say:

“The agent does not blindly persist. It first runs a stateless review so the user can validate the calculation before saving it.”

0:35 - 0:50 | Persist article

Action:

- click `Create Article`

Show:

- created article summary
- direct links for edit / calculate / table / view

Say:

“Once approved, the reviewed script becomes a reusable CalcsLive article.”

0:50 - 1:10 | Send to Excel

Action:

- open `Bridge to/from Excel`
- click `Send Calc to Excel`

Show:

- Excel populated with metadata block and PQ table

Say:

“The same reusable calculation is now bridged into Excel as a structured, unit-aware worksheet.”

1:10 - 1:35 | Reactive Excel update

Action:

- edit one or more Excel inputs

Show:

- output value updates automatically

Say:

“When spreadsheet values change, the bridge reads the updated PQ table, calls CalcsLive, and writes the new results back into Excel.”

1:35 - 1:55 | Reverse flow from Excel

Action:

- show a compatible Excel-authored table
- click `Get Calc from Excel`

Show:

- reviewed script appears in the app

Say:

“The bridge also works in reverse. A compatible Excel table can be read, converted, reviewed, and persisted as a reusable CalcsLive calculation.”

1:55 - 2:00 | Close

Say:

“This makes CalcsLive a reusable unit-aware calculation engine, with the agent moderating the workflow between cloud logic and spreadsheet execution.”

Supporting Visual Assets

- architecture diagram
- unified app screenshot
- Excel before/after screenshots
- optional GIF of reactive recalculation